

# OS Unit 2 — CPU Scheduling & Synchronization

## Complete Notes

Operating Systems (OS) · Unit 2 · CPU Scheduling & Synchronization · 2024

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### Overview

These notes cover **CPU Scheduling & Synchronization**. How operating systems manage processes, CPU time, memory, storage and devices — and keep it all safe and concurrent. Each section states the core idea, a worked intuition and the points examiners look for.

### 1. Scheduling Criteria

CPU utilisation, throughput, turnaround time, waiting time and response time evaluate schedulers.

**Why it matters:** Scheduling Criteria appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of scheduling criteria in one line.
- A small example showing scheduling criteria in practice.
- Common mistakes students make and how to avoid them.

### 2. FCFS, SJF and Round Robin

FCFS is simple but suffers convoy effects; SJF minimises average waiting time; round robin uses time quanta.

**Why it matters:** FCFS, SJF and Round Robin appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of fcfs, sjf and round robin in one line.
- A small example showing fcfs, sjf and round robin in practice.
- Common mistakes students make and how to avoid them.

### 3. Priority Scheduling

Priority scheduling can starve low-priority processes, which aging solves.

**Why it matters:** Priority Scheduling appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of priority scheduling in one line.
- A small example showing priority scheduling in practice.
- Common mistakes students make and how to avoid them.

## 4. Critical Section Problem

Solutions must provide mutual exclusion, progress and bounded waiting.

**Why it matters:** Critical Section Problem appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of critical section problem in one line.
- A small example showing critical section problem in practice.
- Common mistakes students make and how to avoid them.

## 5. Semaphores

Counting and binary semaphores coordinate processes using wait and signal operations.

**Why it matters:** Semaphores appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of semaphores in one line.
- A small example showing semaphores in practice.
- Common mistakes students make and how to avoid them.

## 6. Classic Synchronization Problems

Producer-consumer, readers-writers and dining philosophers model common concurrency patterns.

**Why it matters:** Classic Synchronization Problems appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of classic synchronization problems in one line.
- A small example showing classic synchronization problems in practice.
- Common mistakes students make and how to avoid them.

## Quick revision

Scheduling Criteria · FCFS, SJF and Round Robin · Priority Scheduling · Critical Section Problem · Semaphores · Classic Synchronization Problems